

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

<think>

Finding Reasoning Paths :

- 1. **Short attention span** → Underlying neurodevelopmental disorder (e.g., attention-deficit/hyperactivity disorder [ADHD], autism spectrum disorder [ASD]) → Behavioral therapy .
- 2. **Difficulty concentrating** → Neurodevelopmental disorder → Behavioral therapy .
- 3. **Hyperactivity and frequently interrupting others** → Hyperactivity → Neurodevelopmental disorder → Behavioral therapy .
- 4. **Acute otitis media** → Rule out hearing impairment (as hearing problems can mimic ADHD symptoms) → No abnormality in hearing .
- 5. **Family history of depression and thyroid disease** → No direct link to current symptoms → Rule out psychiatric or endocrine causes .
- 6. **Impairments in speech and language development** → Rule out syndromic causes (e.g., intellectual disability , neurodegenerative disorders) → No abnormality in speech and language development .

Reasoning Process :

1. **Symptoms Analysis** :

- The child exhibits a short attention span , difficulty concentrating , hyperactivity , frequently interrupting others , and a forgetful nature . These are hallmark features of **ADHD** , particularly the combined presentation (hyperactive - impulsive and inattentive symptoms) .
- The boy 's behavior is consistent with inattentiveness and impulsivity rather than hyperactivity alone , as the teachers report that he is " more active " rather than restlessly moving or showing excessive physical energy .

2. **Medical History** :

- The child has no history of seizures , febrile seizures , or recurrent infections that could suggest an organic cause for his symptoms .
- The history of **acute otitis media** was treated appropriately , and examination shows no abnormalities , ruling out hearing impairment as a contributor to his symptoms .

3. **Developmental and Behavioral Evaluation** :

- The child has met all developmental milestones and has no abnormalities in speech and language development . This makes syndromic or structural neurological causes (e.g., syndromes like Prader-Willi syndrome , genetic neurodegenerative disorders) less likely .
- There is no evidence of motor dysfunction or intellectual disability based on the description of his symptoms and normal development .

4. **Family History** :

- The mother has major depressive disorder , but there is no indication of a genetic predisposition to depression in this case .
- The father has Graves ' disease , but there is no evidence of thyroid dysfunction or psychiatric symptoms in the child .

5. **Differential Diagnosis** :

- **ADHD**: The most likely diagnosis given the combination of inattentiveness , hyperactivity , impulsivity , and normal developmental milestones .
- **Autism Spectrum Disorder (ASD)**: While some symptoms overlap (e.g., difficulty concentrating , hyperactivity), the absence of social communication deficits and impaired speech and language development make ASD less likely .
- **Neurodevelopmental and Psychiatric Disorders**: Other syndromic or psychiatric conditions (e.g., attention-deficit/eating disorder , Rett syndrome) are unlikely given the normal developmental history and lack of associated medical findings .

6. **Treatment Considerations** :

- ADHD is a behavioral disorder , and first-line treatment typically involves **behavioral therapy** (e.g., cognitive-behavioral therapy [CBT] or parent training) in combination with medications (e.g., methylphenidate , non-stimulant options) if behavioral modifications are insufficient .
- The absence of organic causes